

General Risk Assessment

This form allows for assessment of the hazards and risks associated with projects involving the use of **biological agents, genetically modified microorganisms**, or of materials such as **cells and tissues** that may be contaminated with these agents. It also identifies those at risk from the work and the measures necessary for preventing or controlling these risks.

The work described within this risk assessment **must not** start until final approval has been given.

This risk assessment must be amended with any changes to the work, at which point a new approval must be given. Changes to the people registered on this project should also be recorded.

Project Details

Activity title: Characterisation of biomaterials as Tribo Electric Nano Generators (TENGs)

Faculty: Faculty of Engineering

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Section/Division: -

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Registered Person	Date Added	Remove User?
Imogen Young	22 May, 2026 15:00:02	No
Rebecca Stewart	22 May, 2026 15:00:02	No
Maya Thabal Herron	22 May, 2026 15:20:59	No

Core Information

Background and aims of the project:

A tribo testing rig has been built to be able to characterise a series of biomaterials for their performance as TENGs

Equipment: Prusa MK3s Printer (modified as explained below), DC Power Supply GPC-3030D, two Keithley DMM6500s, custom PCB, load cell, 3D printed and laser cut parts, M3 nuts and bolts (of various lengths)

Campus:

SOUTH KENSINGTON
Building:
DYSON
Room:
B01
If parts of this project are taking place in different rooms, which are not part of a facility listed above, indicate which part(s) of the project will be conducted in each of these rooms.
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How often do you estimate the activity will take place?
Daily
Please give details
-

Hazards									
Hazards Summary	Hazard Description	Hazards	Likelihood	Severity	Risk Level	Risk Controls	Likelihood	Severity	Risk Level
-	-	-	-	-	-	-	-	-	-
Action(s)									
Action Title		Action Description		Action Owner		Action Assignee(s)		Action Due Date	
Mechanical	The system is delivering forces of up to 30N repeatedly. In addition all 3 axis move at relatively fast speeds. There is no casing or sheilding around the system. The only modifications to the Prusa were to remove the hot end of the print head, and replace with a 3D printed attachment	Any person who is seting up tests or working in the immediate area could get their hand caught in the machine and get a crush or pinch injury.	Likely	Moderate	High	There is a two step system to begin motion of the rig meaning it is less likely to have the rig moving while your hands are still within the mechanisms. In addition, there is a safety programme which will stop the system and raise the print head if the force ever	Unlikely	Moderate	Medium

	which functions as a flat 4mm square surface which is used to apply the force to a material. In addition, the standard print bed has been removed and replaced with a Laser cut and 3D printed platform which houses the load cell. This shouldn't pose any new hazards and removes the main hazard of the hot end of a 3D printer.					exceeds 35N. The printer will also be switched off when changing the material under investigation as that is the only time hands are meant to be within the rig and are at risk. The control programme is constantly reading the force measurement from the connected load cell as a background loop. There is a safety loop which is constantly monitoring the force			
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						reading, and if that force reading is to ever exceed 35N, it sends a G-code command to the printer immediately to stop current motion, raise 10cm, then emergency stop which locks the printer from receiving any more commands or moving any more. This has been tested by manually and safely adding weights to the system until it exceeds 35N, at which point the Prusa raises			
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						the print head and then locks itself and the code correctly raises a safety error. If this were to fail however, The printer can be turned off on the switch at the side, or directly unplug the power cable which unlocks the axis allowing an individual to manually move the print head to a safe position.			
Electrical	The TENGs could in theory produce	Anyone working in the immediate	Possible	Moderate	Medium	The wires are managed with cable ties and the	Unlikely	Moderate	Medium

	voltage spikes of up to - + 12V (which is capped by the amplification of the PCB). In addition, current spikes could be produced of up to 1mA (0.001A). There are exposed electrodes and some exposed points for connectors on the PCB.	area could touch something and recieve an electrical shock.				whole system is placed out of the way where it is less likely to be touched accidentally. In addition the OPAMPs will not allow more than +- 12V on the output pins which avoids more dangerous voltage spikes. In addition all systems are turned off when testing is not occuring. During testing, all pins that may provide a voltage spike are covered			
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						by the cables connected to them thus removing possibility of accidental touching. In addition, the +-12V inputs are covered by the insulation of the screw on banana plug terminals, and the base of those terminals cannot be reached while the PCB is on the raised mount. There are no other exposed contact points during testing and so an enclosure is not needed.			
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Waste Management
Waste Disposal - Types of waste generated
General
Provide details of how you are going to dispose of each type of waste selected
The only expected waste would be fabric offcuts which will be placed in a general waste bin in the basement. The test rig is expected to stay in the lab and be used for Tribo Testing on an ongoing PhD project (Maya's research). If the rig is no longer being used, the power supply, Keithley and associated cables will be returned to the general lab store. The laser cut and 3D printed parts which have been added to the Prusa will be taken off and stored alongside the PCB in the e-body Lab archive in case they are required in the future. The Prusa will go back to the general lab storage where it was acquired from.
Emergency Arrangements
Is lone working permitted for this activity?
No
Provide confirmation that lone working has been authorised
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Fire Procedure
In the event of a fire, all systems are turned off by unplugging the 3 cables (one for the printer, one for the power supply and one for the Keithley) and then the evacuation protocol is followed.
Spillages Procedure
There are no drinks allowed in the lab and no chemicals being used in this area.
First Aid Procedure
All machines will be turned off via unplugging their power cables. The first aid kit located by the sinks in B01 will be used if required, and the contact list there will be used to communicate with the designated First Aiders if assistance is needed. If the accident is severe, campus security will be called.